



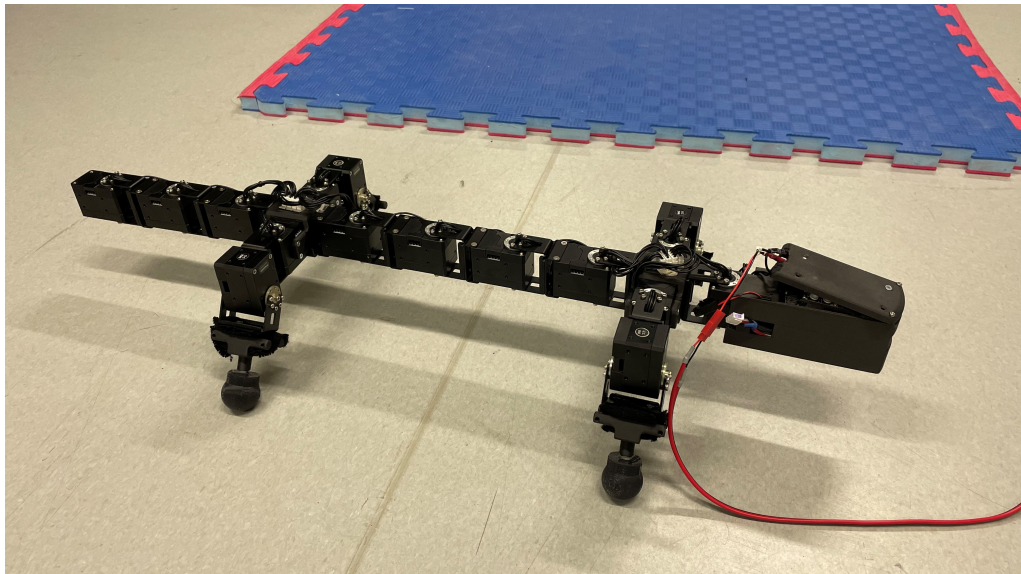
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CURRENT BASED CONTACT ESTIMATION FOR LEGGED ROBOTS

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CHAPTER 1

INTRODUCTION

Chapter 1 explains why contact estimation is important for legged robot and presents the advantages of calculating this information based on currents. Then, some important concepts for this projects will be introduced. Finally, we will present the objective of this project.

1.1 MOTIVATION

In case of real world scenario like disaster recovery, legged robots can navigate in challenging and complex environments that are non-flat [1]. Legged robots take advantage in situation where wheeled robots cannot move [2]. To achieve stable-legged locomotion, some sensory information like ground reaction forces (GRFs) are needed. Therefore, many legged robots install force or pressure sensors on the feet to measure ground reaction forces. However, frequent contact with the ground tends to damage the sensors on the feet, reducing reliability and durability of the robot [3]. Using current based estimation to compute the ground reaction forces is an alternative method that do not need specialized sensors that should resist to the impact forces during locomotion, and respect a desired form factor to be incorporated to the robot.

1.2 BACKGROUND

This section gives an overview of the main concepts used in this project. The different methods to compute ground reaction forces will be discussed. Then we will briefly introduce machine learning concepts and explain the importance of feature engineering.

1.2.1 GROUND REACTION FORCES

Ground reaction forces (GRFs) refer to the forces that are transmitted through the feet to the ground during physical activities such as walking, running, jumping, and more. They are used in the robotics field to generate and adjust the gait in response to different environments and tasks.

Lee and Goswami [4] used ground reaction forces to maintain balance on humanoid robots. Their method is based on momentum. By controlling the desired ground reaction force and center of pressure at each support foot, their method can naturally deal with non-level and non-stationary ground at each foot-ground contact, and can adapt to different frictional properties.

Focchi et al. [1] proposed a planning/control framework for quasi-static walking of quadrupedal robots, implemented for a demanding application in which regulation of ground reaction forces is crucial. Experimental results demonstrate that their 75-kg quadruped robot is able to walk inside two high-slope

(50°) V-shaped walls. Hutter et al. [5] used a contact-force optimization method to achieve static walking on a surface with approximately 40° inclination. Their proposed control strategy allows for a hierarchical task decomposition while simultaneously regulating the inner forces between the contact points. They could significantly lower slippage when activating the contact force alignment. There is no need to further develop why ground reaction forces are used. The question now is how they are computed.

1.2.2 SENSORS FOR GRFS

Here are some methods to compute ground reaction forces. Liu et al. [6] developed a mobile force plate and 3-D motion analysis system to measure triaxial ground reaction forces (GRFs) and 3-D orientations of feet. The developed system is made for human gait analysis. The results obtained show a good correspondence between the measurements of the newly developed system and the reference devices. Nishiwaki et al. [7] developed a six-axis force sensor to measure the ground reaction force of humanoid robot. Jumping and kicking motions were carried out using the wearing type sensor. They also succeeded to realize stable walking on a humanoid using the six-axis force sensor information. However, it would be difficult to adapt this kind of system to a quadruped robot for example, since most of them have do not have a human-like foot on which we could fix the sensor. The methods presented above are designed for biped robots. The possibility of tumbling is higher than for any other multi-legged systems.

For quadruped robots, ground reaction forces can improve walking on non-flat or slippery terrains. Hutter et al. [8] developed a highly mobile and dynamic quadrupedal robot called ANYmal. The robot is designed with a focus on outdoor suitability, simple maintenance, and user-friendly handling to enable future operation in real world scenarios. Contact estimation is achieved with Optoforce sensors that are a kind of optical force sensors. Getting contact estimation with sensors fixed under the feet of a robot can be an issue because adding more sensors is more expensive. Moreover adding more components increases the risk of failure. For all these reasons, it is better to have information directly from inside the body if possible. For instance by using the motor currents to calculate the ground reaction forces. This could be done with machine learning techniques.

1.2.3 MACHINE LEARNING

Machine learning algorithms can be divided into five main groups that are supervised learning, unsupervised learning, semi-supervised learning, reinforcement learning and deep learning as illustrated in Figure 1.1. Supervised learning algorithms try to learn a function that maps an input (features) to an output (target). It is possible to predict the output values for new data based on the model obtained with the training data. Unsupervised machine learning algorithms, unlike supervised machine learning algorithms, do not rely on labeled data. It means that the algorithm must find patterns or relationships in the data without any guidance [9]. In reinforcement learning, an agent learns to interact with its environment in order to maximize a reward. The agent learns by trial and error, using feedback from its environment to adjust its strategy and improve its performance over time.

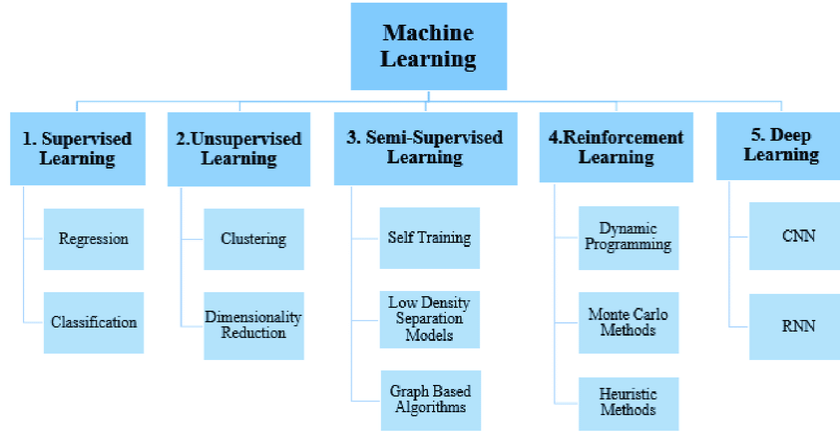


FIGURE 1.1
Overview of machine learning types and methods (Nassif et al., 2019 [10]).

For this project, we will be using supervised learning. As shown in Figure 1.2, the data must be split into training and testing dataset. The model is build with training data and evaluated with the testing data to achieve classification or prediction [9]. It is crucial that the test set is used only for estimating a final, unbiased assessment of the model's performance.

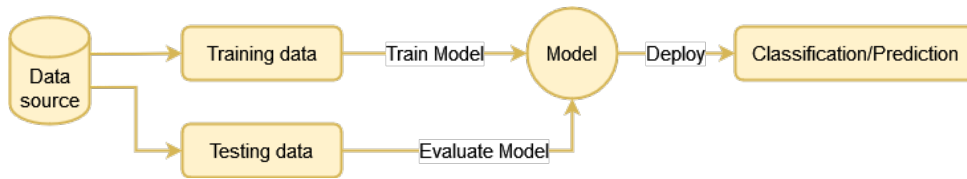


FIGURE 1.2
Supervised learning workflow

In Figure 1.2, between the training data and the model step, we can split the training data into two subset to improve performance of the model by using the k-fold cross validation method. This method will be explained in more details.

K-FOLD CROSS VALIDATION

To improve the performance of the model, we can separate the training dataset in a training set and a validation set (see Figure 1.3). The validation set gives some metrics performance to evaluate the model. We repeat this step k times, and at each step a different group stands for the validation set. After k steps, the final model that fits the dataset best is produced, ready to be tested with the testing dataset. This method is called k-fold cross validation.

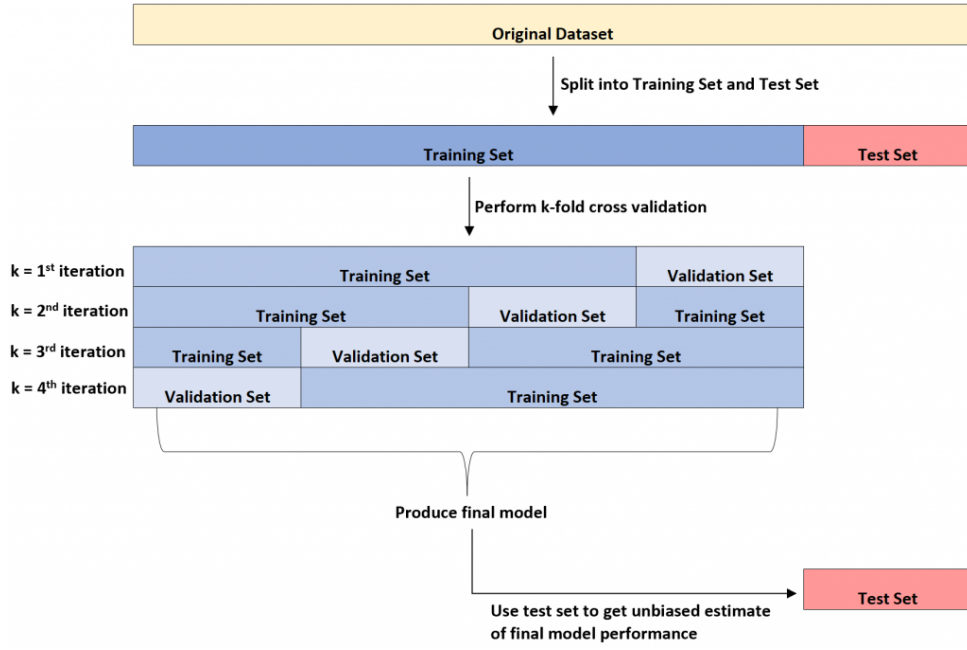


FIGURE 1.3
k-fold cross validation (Zach, 2021 [11])

The main types of algorithms to achieve supervised learning are classification and regression. Classification is a predictive model that approximate discrete output variables from input variables. Regression algorithms predict a continuous value based on the input variable. In this project, classification would lead to contact based estimation and linear regression would conduct to ground reaction forces prediction. It is more accurate to predict continuous values than discrete values if the prediction is used for gait analysis.

MODEL EVALUATION METRICS

Botchkarev et al. [12] classify the mean square error (MSE)/root mean square error (RMSE) and the mean absolute error (MAE) as the most popular metrics.

The mean absolute error represents the average of the absolute difference between the actual and predicted values in the dataset. It measures the average of the residuals in the dataset [13].

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}| \quad (1.1)$$

Mean squared error represents the average of the squared difference between the original and predicted values in the data set. It measures the variance of the residuals [13].

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y})^2 \quad (1.2)$$

Root mean squared error is the square root of Mean Squared error. It measures the standard deviation of residuals [13].

$$RMSE = \frac{1}{N} \sqrt{\sum_{i=1}^N (y_i - \hat{y})^2} \quad (1.3)$$

The coefficient of determination or R-squared represents the proportion of the variance in the dependent variable which is explained by the linear regression model. It is a scale-free score i.e. irrespective of the values being small or large, the value of R square will be less than one [13].

$$R^2 = 1 - \sum \frac{(y_i - \hat{y})^2}{(y_i - \bar{y})^2} \quad (1.4)$$

1.2.4 FEATURE ENGINEERING

Feature engineering is the process of creating representations of data that increase the effectiveness of a model [14]. This can involve creating new features by combining or modifying existing features, or selecting a subset of the most relevant features from a larger set. Figure 1.4 shows the steps to do to achieve machine learning. We can see that feature engineering is a big part of it.

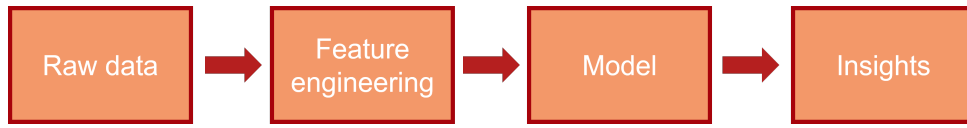


FIGURE 1.4
Feature engineering in machine learning.

Feature engineering in machine learning contains mainly four processes: feature creation, transformations, feature extraction, and feature selection. Feature creation involve generating new data, feature extraction is mainly used to reduce the volume of the data. Feature extraction consists in identifying useful information in the dataset. In this project, we used very simple notion of feature engineering.

“ You can have data without information, but you cannot have information without data. ”

Daniel Keys Moran

Now that we have presented the background knowledge to understand what has been done in this project, it would be good to recall the objectives of the project.

1.3 OBJECTIVE

This project aims at exploring the relation between feedback current in the motors of a quadruped robot and the ground reaction forces. This will be achieved with supervised learning algorithms such as simple linear regression and multiple linear regression. The final objective is deducing the ground reaction forces in Fz directly from the motors current.

CHAPTER 2

MATERIAL AND METHODS

In this chapter, we will explain how we collected the data with the robot (polymander) and with the force plate. The protocol of experiments will be presented. The second part concerns the feature engineering process. We will explain how we select the data and how we extract the relevant features. Finally, the algorithms used to achieve supervised learning will be described.

2.1 DATA COLLECTION

To do supervised learning, we need currents and ground reaction forces. The current are extracted from a robot called polymander and the ground reaction forces are measured with a force plate.

2.1.1 POLYMANDER

The robot used during the experiments to collect the data is a quadruped robot developed in BioRob Laboratory by Laura Paez. The robot is called polymander and is composed of 16 Dynamixel servomotors. They are distributed as follows: eight motors for the spine and two motors per limb as shown in Figure 2.1.

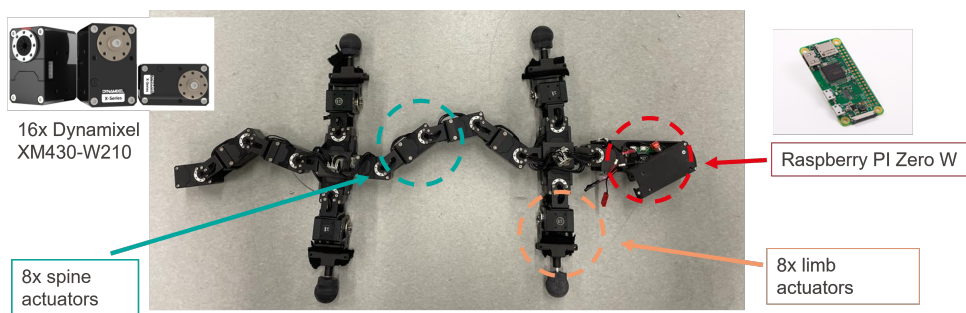


FIGURE 2.1

Polymander is composed of 16 actuators (8 for the spine and 2 per limb). The actuators are XM430-W210 servo motors. The computer is a Raspberry PI Zero W.

The computer in the head of the robot is a Raspberry PI Zero W. It is composed of a 1GHz, single-core CPU and a 512MB RAM. There are three ways to communicate with the robot: micro-USB and mini-HDMI port, 802.11n wireless LAN or Bluetooth 4.0. The one we use for the experiments is the wireless LAN. The Dynamixel XM430-W210 robot servos are adapted both to articulated robots and

radio-controlled vehicles. It is built with fully integrated DC Motor, controller, driver, sensor, reduction gear, network in one DC servo module. There are six operating modes including PWM, velocity, torque, current and trajectory controls.

The code in the robot is in C++. There is a walking gait based on position control implemented. The sensors in the motors return feedback like position, velocity, current, temperature, input voltage, etc. This information can be access with the control table.

Now that we have information about the robot, we have to find a way to measure the true value of ground reaction forces. To do this, we will use a force plate.

2.1.2 FORCE PLATE

To calculate the true value of the ground reaction forces, three options emerged.

- Kistler Force Plate
- BOTA Rokubi F/T Sensor
- ATI Mini45 F/T Sensor

The first option is the Kistler Force Plate system already implemented in the GAIT Laboratory at EPFL. Kistler provides a Data Acquisition System (DAQ) for the signal processing of the force plate. An USB 2.0 connection allows to connect the laptop to the DAQ System and control the force plate with the BioWare software (see Figure 2.2).

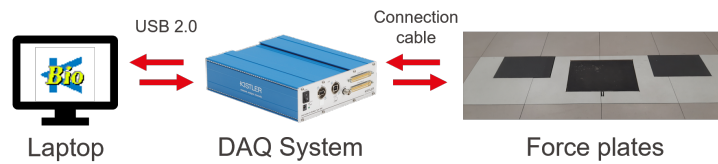


FIGURE 2.2
Kistler Force Plate setup and communication [15].

Kistler force plates are designed for human gait analysis, and since polymander weights 1.8 kg, we had to check if the signal to noise ratio was not too high. The other option is to build a custom force plate with a 6 axis force torque sensor like the BOTA Rokubi F/T or the ATI Mini45. These two solutions are a backup if the results with the Kistler force plate are not satisfying.

2.1.3 PROTOCOL OF EXPERIMENTS

The dataset is composed of 45 experiments made on the front left leg. We made the learning more robust with amplitude and frequency modulation of the encoded pattern. The amplitude vary from 0.2, 0.35 and 0.5 rad and the frequencies vary from 0.1, 0.5 and 1.0 Hz. For each scenario, we repeated the experiments five times (see Table 2.1) to have a sufficient amount of data. For frequency $f = 0.5$ Hz, we record data during 60 seconds. For frequency $f = 0.1$ Hz, the robot makes steps slower so we record data during 90 seconds and for frequency $f = 1$ Hz, we record data during 30 seconds.

TABLE 2.1

Modulation of the amplitude and frequency. Each scenario is repeated 5 times.

		Frequency		
		0.1 Hz	0.5 Hz	1.0 Hz
Amplitude	0.2 rad	5 exps	5 exps	5 exps
	0.35 rad	5 exps	5 exps	5 exps
	0.5 rad	5 exps	5 exps	5 exps

An animal gait based on a position controller was already implemented in the robot. We adapted the gait to make the robot move only one leg at a time. Having the robot remaining over the force plate allows more repetitions of the same movement compared to when the robot is walking and make only three steps on the force plate. This increases the dataset size and allows more stability when recording. We encountered a hardware problem when moving only one leg. Polymander cannot keep balance, thus the limb does not leave the ground and the robot lean towards the ground. We tackled this problem by moving the center of gravity the opposite way of the lifting limb. We can see the offset of the motors in Figure 2.3.

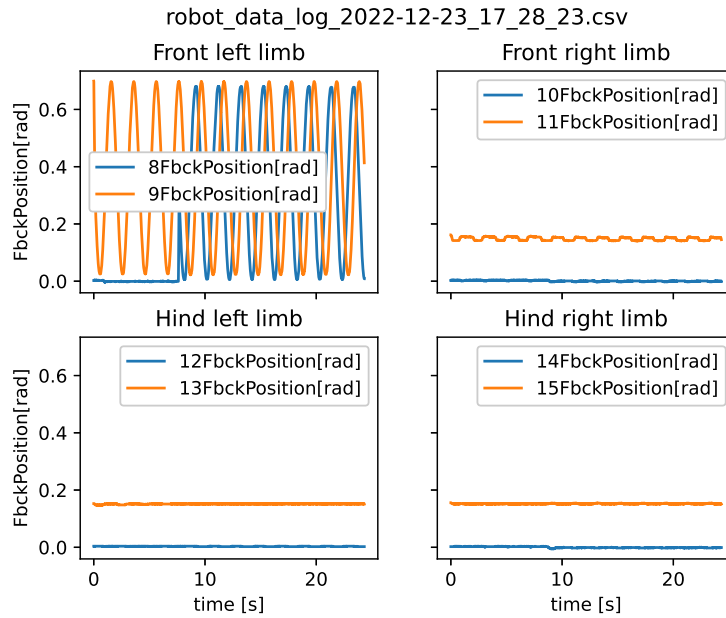


FIGURE 2.3

Feedback position of motors. We can see that to keep balance when moving one limb, there is an offset applied to the other motors.

For each experiment, we placed the robot above the force plate, with the limb raised and not touching the force plate. Then we press enter to activate the force plate recording and right after that we start the robot. The robot begins each experiments with an initial sequence consisting in making four steps on the force plate with an amplitude of 0.35 rad. This allows to align the signal of the force plate and the robot's motors. Further explanation is given in subsection 2.2.2.

2.2 FEATURE ENGINEERING

In this part, the data processing will be presented. First the different steps to get rid of the noise in the signal. Then, we tackle the problem of the delay between the robot motors sensors and the force plate.

2.2.1 SIGNAL PROCESSING

The first thing to do is filtering the noise of the feedback currents recorded by the motors sensors and the forces recorded by the force plate. For both signal, we remove the noise with a Gaussian filter as shown in Figure 2.4. Different values for the σ parameter have been tested and by visually looking at Figure 2.4a and 2.4b, we chose a value of $\sigma = 10$ for the feedback currents and a value of $\sigma = 20$ to filter the force Fz.

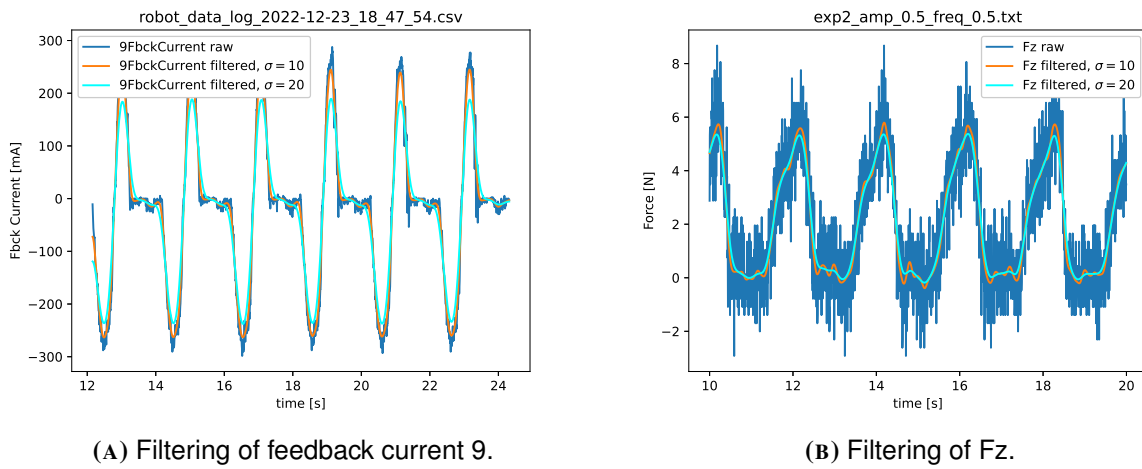


FIGURE 2.4

Remove the noise in signal with a Gaussian filter ($\sigma = 10$ and $\sigma = 20$)

The next step is resampling the force plate signal based on the robot frequency. The sampling rate of the robot is around 166 Hz and the sampling rate of the force plate has been chosen at 200 Hz. After that both signals of polymander motors and of force plate are ready to be treated to manage the delay between sensors.

2.2.2 MANAGING THE DELAY BETWEEN SENSORS

The motor currents are measured with integrated motor sensors and forces are measured with a force plate. This means that there are different computers, different clocks, etc. running at the same time. Moreover, there is human interaction to start the robot and the force plate. All these parameters generate a delay between the current signal and the force signal. When the delay was not treated, there was no correlation at all between the motor currents and the forces. Results began to appear only when we took care of the delay between the different sensors.

Managing the delay between the force plate and the robot's recording data to ensure alignment of the two signals is necessary, since we want to achieve supervised learning. To do so, we defined an initial sequence the robot could make at the beginning of each experiment. This consists of four steps with a specific amplitude of 0.35 rad as shown in Figure 2.5. When the feedback position of calf motor is at a minimum and the Fz force is at a maximum, we know that the limb touch the ground.

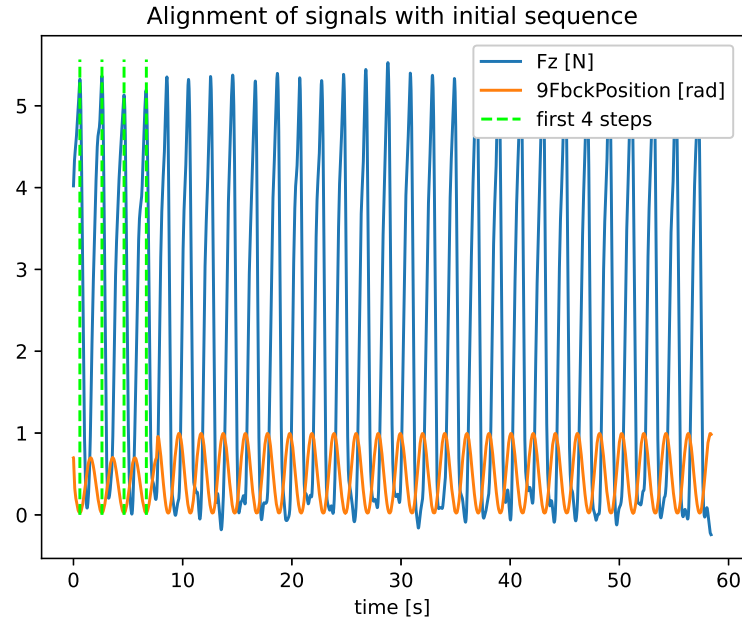


FIGURE 2.5

Alignment of force plate signal (Fz) and polymander signal by detecting when the limb touches the ground (minima of limb fbck position and maxima of Fz).

Now that we aligned motor current and forces signals, the next step consists in removing the initial sequence to keep the window with the chosen frequency and amplitude. Figure 2.6 shows the feedback currents and the force Fz ready to be use to the supervised machine learning.

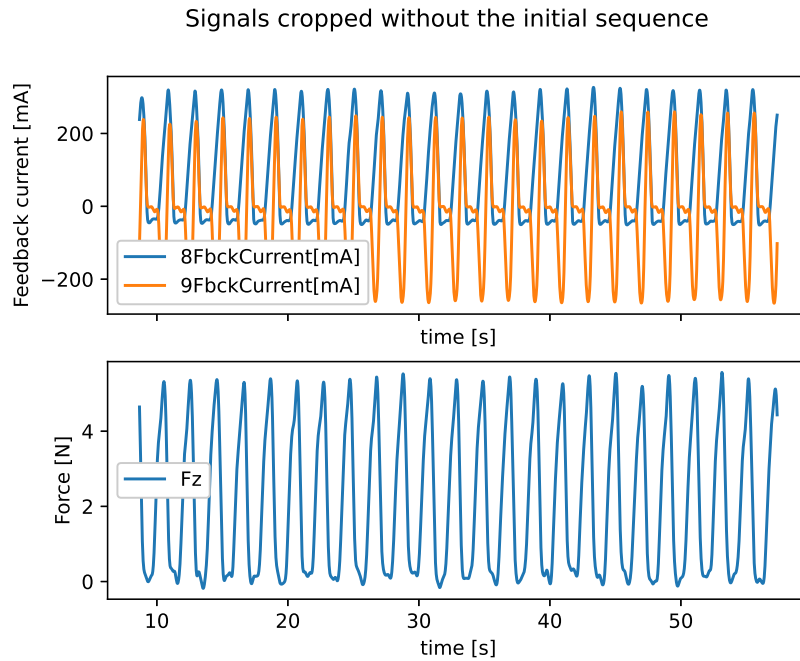


FIGURE 2.6

Signals of feedback current 8 and 9 and force Fz perfectly aligned. The initial sequence has been removed. All signals are filtered and have the same frequency.

2.3 SUPERVISED LEARNING

The steps to build a model with supervised machine learning are shown in Figure 1.4. The training data are the feedback currents in the limb that is moving (hip motor and calf motor currents) and the normal force measured by the force plate (Fz), see Figure 2.6. The feature engineering part is explained in section 2.2. The algorithm used to train the model is regression based.

To do the learning part, we use scikit-learn library of python. We will use functions to split the data set, to predict a continuous value with regression and some evaluation metrics are implemented. Before training the model, the dataset must be split in a training set and a testing set. To begin with, we build the model and made the prediction with a simple linear regression. Then we increased the complexity by using multiple linear regression.

TABLE 2.2

Variables used to calculate the simple and the multiple linear regression. Feedback current 8 and 9 are the independent variables and Fz is the dependent variable.

Time [s]	Hip motor current [mA]	Calf motor current [mA]	Fz [N]
t_1	Hip motor current ₁	Calf motor current ₁	Fz ₁
t_2	Hip motor current ₂	Calf motor current ₂	Fz ₂
...	...	...	...
t_n	Hip motor current _n	Calf motor current _n	Fz _n

The metrics used to evaluate the models performance are the R-Squared, Mean Square Error (MSE)/Root Mean Square Error (RMSE) and Mean Absolute Error (MAE).

CHAPTER 3

RESULTS

This chapter shows the results obtained on the 45 experiments. First, we will compare the model obtained with a simple linear regression and multiple linear regression. Then, the influence of the amplitude and the frequency on the results will be analyzed by comparing these regression metrics: R-Squared, Mean Square Error (MSE)/Root Mean Square Error (RMSE) and Mean Absolute Error (MAE). The amplitude defines the amplitude of the limb movement when the robot is moving and the frequency is the frequency at which it makes each step when walking. In real-life situation, depending on obstacles the robot would encounter, the robot would have to adapt the amplitude of its motion or the frequency of its steps. This is why it is interesting to train the different models to see the impact of amplitude and frequency modulation.

3.1 SIMPLE LINEAR REGRESSION

We first trained a model with simple linear regression. First, with the hip motor current and then with the calf motor current. In Figure 3.1, we can see the model obtained with the linear regression based on the hip motor current. There is a weak correlation between the hip motor current and the force F_z . This can be explained by the motion of the hip that is not directly related with the impact on the ground in the z direction. The prediction matches the frequency, but the model fails to predict the amplitude of the F_z force.

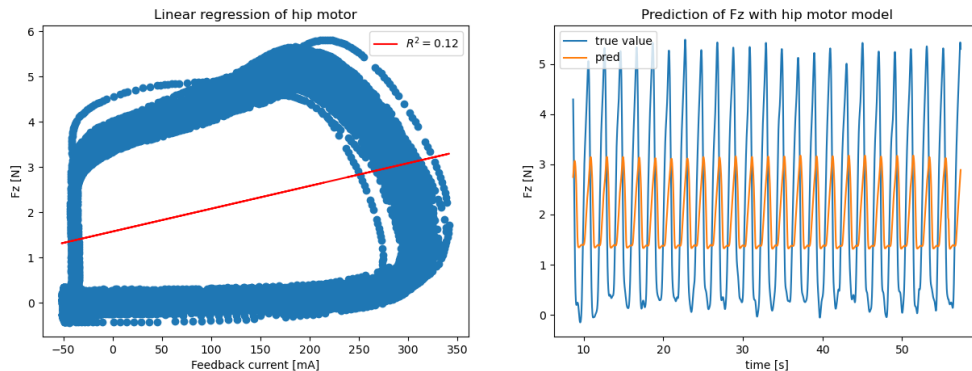


FIGURE 3.1

Simple linear regression with the hip motor current with an amplitude of 0.5 rad and a frequency of 0.5 Hz.

Figure 3.2 shows the model obtained with a simple linear regression based on the calf motor current. We

can see that there is a stronger correlation between the calf motor current and the force F_z than between the hip motor current and the force F_z (see Figure 3.1). However, the prediction far from being accurate enough. There is an offset between the true value and the prediction.

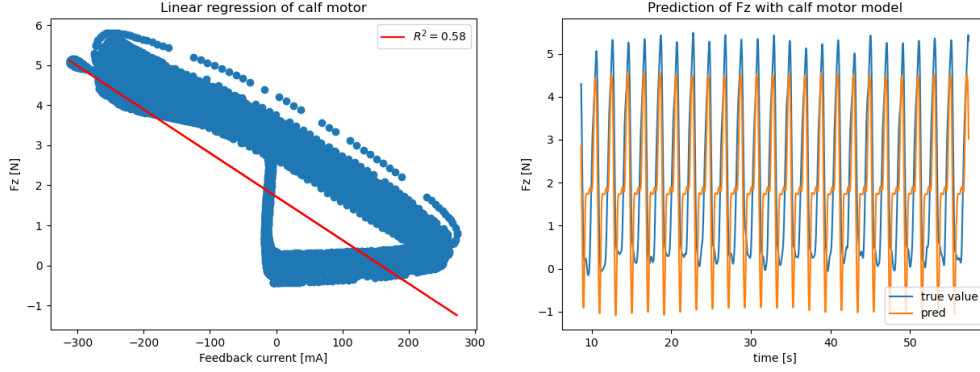


FIGURE 3.2

Simple linear regression with the calf motor current with an amplitude of 0.5 rad and a frequency of 0.5 Hz.

Figures 3.1 and Figure 3.2 allow a better understanding of the role of the hip and the calf motor on the F_z force. The hip motor makes a yaw rotation, making its motion less related to the impact force in the z direction. The calf motor makes a roll rotation, therefore applying more pressure on the force plate. Since these results are not satisfying to predict an accurate value of F_z , we will investigate multiple linear regression.

3.2 MULTIPLE LINEAR REGRESSION

We increase the level of complexity by building the model based on the hip motor current and the calf motor current to see if it is possible to improve the result. Figure 3.3 shows the data in 3D. The force F_z is plotted as a function of hip motor current and calf motor current. The model build with the multiple linear regression is a hyperplane and fit the data with a R-squared of 0.8, which is better than the simple linear regression.

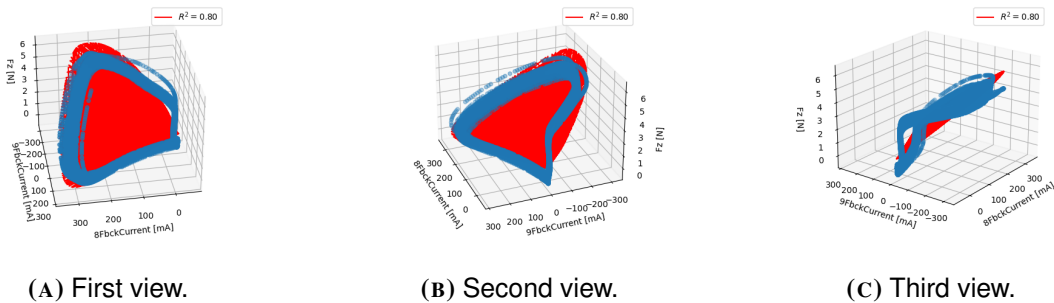


FIGURE 3.3

Representation of data in 3D. The blue points represent the training data. The force F_z is a function of the hip current motor (8) and the calf current motor (9). The red hyperplane represents the model learned by the multiple regression algorithm.

We can see by looking at Figure 3.4 that the prediction of F_z force is improved compared to the prediction made with simple linear regression. However, there is a strange behavior when we look at the force around zero newton. When the force is between 1 N and 5 N, the prediction is more accurate than when the force is between 0 N and 1 N. The fact that there is a high signal to noise ratio in the force plate measurement (see Figure 2.4b) could explain that around zero newton the model is harder to train. This lead to a less accurate prediction.

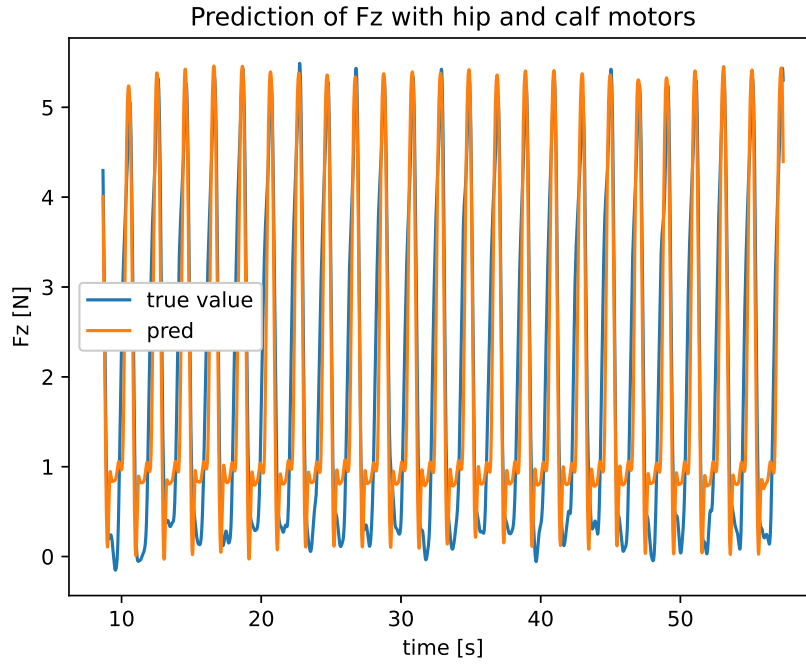


FIGURE 3.4

Prediction of F_z with motor hip and calf motors for an amplitude of 0.5 rad and a frequency of 0.5 Hz.

In this section, we trained the model with normal values of amplitude and frequency (amp. = 0.5 rad, freq. = 0.5 Hz). However, it would be interesting to observe how the model would learn at low and high frequencies, same for amplitude. This will be discussed in sections 3.3, 3.4 and 3.5.

3.3 INFLUENCE OF AMPLITUDE

To observe the impact of the amplitude on the learning, we trained three models, one for each amplitude (0.2 rad, 0.35 rad and 0.5 rad). The R-squared metric, the mean squared error and the root mean squared error help quantifying the performance of the model. We evaluated the metrics for the simple linear regression of the hip, the calf and for the multiple linear regression.

Table 3.1 shows the metrics of hip motor current models. Similarly low values are obtained for the R-squared metric, so it means that the models fit the data badly regardless of the amplitude. The three error metrics show that the error decreases when the amplitude decreases. Therefore, the best predictions are made for an amplitude of 0.2 rad, where errors are smaller.

TABLE 3.1

The R^2 , MSE, RMSE and MAE metrics of simple linear regression with hip motor current model trained at different amplitudes.

Amplitude	R^2	MSE	RMSE	MAE
0.2 [rad]	0.11	1.61	1.27	1.13
0.35 [rad]	0.22	3.00	1.73	1.53
0.5 [rad]	0.15	3.40	1.84	1.62

The results for the linear regression with hip motor current are similar to the calf motor current regarding error metrics. It means smaller for a lower amplitude (0.2 rad).

TABLE 3.2

The R^2 , MSE, RMSE and MAE metrics of simple linear regression with calf motor current model trained at different amplitudes.

Amplitude	R^2	MSE	RMSE	MAE
0.2 [rad]	0.65	0.65	0.81	0.64
0.35 [rad]	0.34	2.54	1.59	1.29
0.5 [rad]	0.48	2.11	1.45	1.26

For all regression models, the predictions are better at lower amplitude. It can be explained by the fact that at lower amplitude the robot makes smaller movements, so the perturbation on the robot are smaller. Only the front left limb is moving, so making bigger movements implies perturbations on the robot. The R-squared metric is very good for multiple linear regression model at Table 3.3

TABLE 3.3

The R^2 , MSE, RMSE and MAE metrics of multiple linear regression model trained at different amplitudes.

Amplitude	R^2	MSE	RMSE	MAE
0.2 [rad]	0.79	0.40	0.63	0.52
0.35 [rad]	0.74	0.99	1.00	0.80
0.5 [rad]	0.75	1.00	1.00	0.78

Now that we analyzed the three model generated for each amplitude for simple and multiple linear regression, let's have a look at the influence of frequency.

3.4 INFLUENCE OF FREQUENCY

To observe the impact of the frequency on the training, we trained three models, one for each frequency (0.1 Hz, 0.5 Hz, 1.0 Hz).

For the hip motor current linear regression models (see Table 3.4), the R-squared metric increases with the frequency, whereas for the calf motor current linear regression models (see Table 3.5), the R-squared metric decreases when the frequency increases. Thus, hip motor current allows a better fit of the data at high frequency, and calf motor current allows a better fit of the data at low frequency. A possible explanation would be that possibly at high frequency, the hip motor have a bigger impact on the force applied to the ground due to the inertia.

TABLE 3.4

The R^2 , MSE, RMSE and MAE metrics of simple linear regression hip motor model trained at different frequencies.

Frequency	R^2	MSE	RMSE	MAE
0.1 [Hz]	0.08	3.72	1.93	1.70
0.5 [Hz]	0.13	2.78	1.67	1.49
1.0 [Hz]	0.49	1.05	1.02	0.89

TABLE 3.5

The R^2 , MSE, RMSE and MAE metrics of simple linear regression calf motor model trained at different frequencies.

Frequency	R^2	MSE	RMSE	MAE
0.1 [Hz]	0.52	1.88	1.37	1.09
0.5 [Hz]	0.45	1.75	1.32	1.06
1.0 [Hz]	0.17	1.77	1.33	1.14

The R-squared metric for multiple linear regression is higher at high frequency as shown in Table 3.6. However, the difference is not as high as for simple linear regression. So models have all a good fit of the data for all frequencies. The error metrics (MSE, RMSE and MAE) all become smaller when the frequency increases. This mean that the prediction is better at high frequency.

TABLE 3.6

The R^2 , MSE, RMSE and MAE metrics of multiple linear regression model trained at different frequencies.

Frequency	R^2	MSE	RMSE	MAE
0.1 [Hz]	0.66	1.35	1.16	0.96
0.5 [Hz]	0.67	1.05	1.03	0.82
1.0 [Hz]	0.80	0.42	0.65	0.53

Now that we have discussed the influence of amplitude independently from the frequency (see section 3.3) and the influence of the frequency regardless of the amplitude (see section 3.4), we will analyze the metrics of linear regression with a different approach. It is interesting to observe what are the impact at each amplitudes and frequencies combination.

3.5 INFLUENCE OF AMPLITUDE AND FREQUENCY ON METRICS

In this section, we will focus on the metrics of the multiple linear regression only because it is the most promising regression technique for this project. The results are systematically better with the multiple linear regression models than with the simple linear regression models.

We can see in Figure 3.5 that the metrics for all amplitudes and all frequencies are between 0.6 and 0.8. Knowing that 1 is the maximal value of R-squared, we can affirm that the models have all a good fitting of the data during the training part. However, R-squared metrics is not sufficient to evaluate the performance of a model. The error metrics allow a better understanding of the models performance by evaluating the model's predictions.

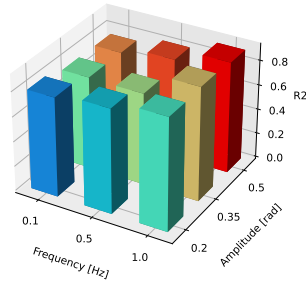
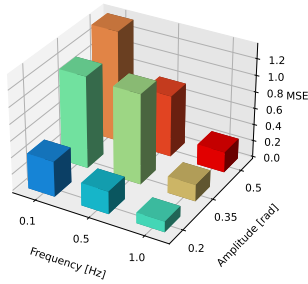


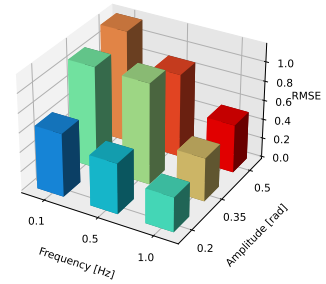
FIGURE 3.5

Evaluation of MLR models with R-squared metric as a function of frequency and amplitude.

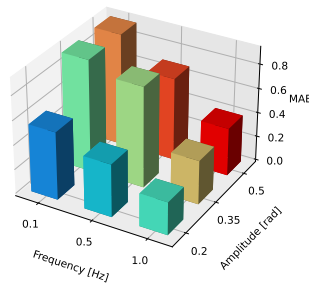
MSE, RMSE and MAE converge to the same ranking of the nine models produced by modulation of frequency and amplitude as illustrated in Figure 3.6. That is errors decrease at higher frequency and lower amplitude. The worst model's prediction are obtained at amplitude of 0.5 rad and frequency of 0.1 Hz. This is consistent with the conclusions reached in sections 3.3 and 3.4. We can see that the RMSE and MAE are more strict on the models that obtained the best performance with MSE.



(A) MSE metric.



(B) RMSE metric.



(C) MAE metric.

FIGURE 3.6

Evaluation of MLR models with Mean Squared Error (MSE), Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) as a function of frequency and amplitude.

CHAPTER 4

DISCUSSION

Results obtained with simple linear regression have shown that models based on calf motor current make better prediction than hip motor current. This makes sense since in the way the robot is walking (animal-like gait), the calf actuator makes a roll rotation that induces the weight on the ground whereas the hip actuator makes a yaw rotation and therefore is less related to force in z direction. What could be interesting is thinking about the fact that the hip motor current based model would lead to better prediction for the ground reaction force in the x direction. Thus, if we extend our model to predict ground reaction forces in x, y and z direction, it would be wise to keep in mind that hip motor (yaw rotation) would achieve better performance to predict F_x and calf actuator (roll rotation) to predict F_y and F_z . With axis x being the one in which the robot moves forward.

All model evaluation metrics have shown that multiple linear regression models performance was better than simple linear regression. This show that even if a motor by itself is not directly related to the output of the model (hip motor), it can still help the other motor to improve the prediction. What has been discussed in the last paragraph could also be applied to multiple linear regression. For example by weighting the data of the motors to give more importance to the hip motor for F_x prediction and more weight to the calf motor for F_y and F_z prediction.

The fact that we obtained smaller error metrics for a lower frequency is not positive, since in real-life situation, the robot would need more to do motion with big amplitude than with low amplitude.

There is one point that can be discussed for the measures during experiments, it is the data measured by the force plate. The noise signal to noise ratio is very high. We had to filter the signal with a high sigma. This is problematic since force F_z represents the true value and should normally be precise.

CHAPTER 5

FUTURE WORK

Most of the work in this project is oriented on the data collection aspect (protocol of experiment), the signal processing part and the analysis of very simple regression models. We have obtained encouraging results indicating that this work deserves further study. There are many things that can be improved to obtain better model performance and therefore better prediction. First of all, the feature engineering part can be improved by exploring more complex techniques that we have not used in this project. Moreover, there is one important thing that was not implemented in the learning process due to a lack of time. This is the k-fold cross validation. This technique prevents overfitting and helps improving model prediction.

The regression techniques used to train the models and predict the resulting force are very basic: simple linear regression and multiple linear regression. However, other methods of supervised learning would deserve to be further investigated, like polynomial regression for example. Another aspect of the supervised learning methods that can be explored is the classification algorithms. It would imply predicting if the foot of the robot is touching the ground or not. This technique would return less precise information than the ground reaction forces, but perhaps we could train more robust models.

In this project, we focused our work on moving only one limb at a time because we had to check if it was possible or not. However, now that results have been obtained with one limb, it would be possible to implement the same thing on a walking robot and to use all available currents to train the model. Perhaps spine motors could improve the model's performance even if they are not directly related to the ground reaction forces.

Exploring better ground true force measurement devices more adapted to robots than a force plate initially designed for human could also be something to improve.

CHAPTER 6

CONCLUSION

Predicting the ground reaction forces with simple supervised learning technique like linear regression is possible. With hip and calf motor currents of one limb as independent variables, we were able to predict the ground reaction force in z direction. We trained different models by modulating the amplitude and the frequency to explore the impact of these variables on the model's performance.

We have to keep in mind that results obtained with simple linear regression have shown that models based on calf motor current make better prediction than hip motor current. This opens up new perspectives, where by analyzing the conception of the robot, we could weight the importance of each motors depending on which limb we want to predict the ground reaction force and on which force is predicted (F_x , F_y , F_z).

The regression techniques used to train the models and predict the resulting force are very basic. However, other methods of supervised learning would deserve to be further investigated, like polynomial regression for example. Another aspect of the supervised learning methods that can be explored is the classification algorithms. In this project, we focused our work on moving only one limb at a time because we had to check if it was possible or not. However, now that results have been obtained with one limb, it would be possible to implement the same thing on a walking robot and to use all available currents to train the model.

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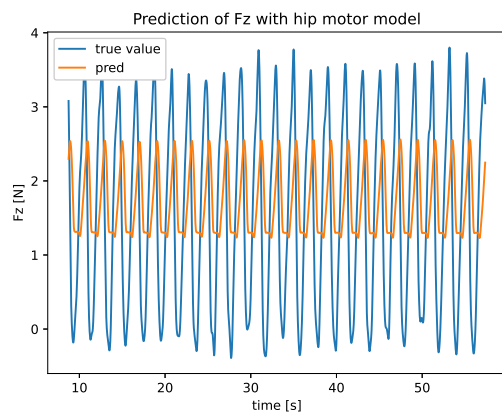
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APPENDIX A

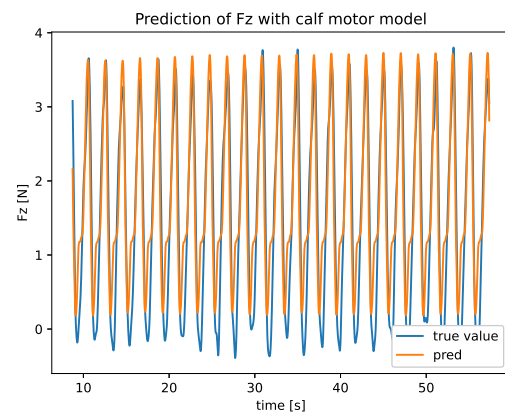
APPENDIX

A.1 INFLUENCE OF AMPLITUDE - PLOTS

A.1.1 AMPLITUDE OF 0.2 RAD



(A) Prediction of Fz with hip motor model.



(B) Prediction of Fz with calf motor model.

FIGURE A.1

Prediction of Fz with simple linear regression model for an amplitude of 0.2 rad.

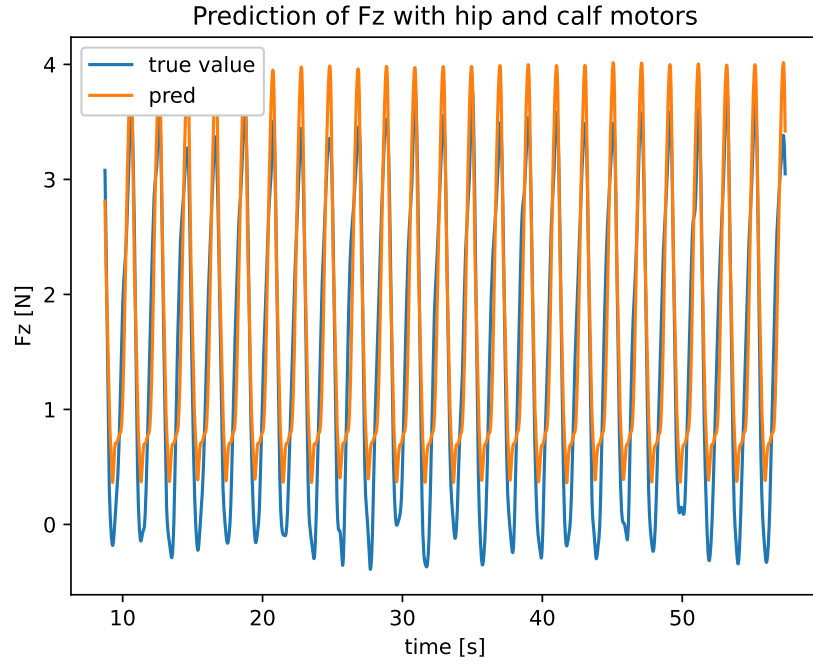
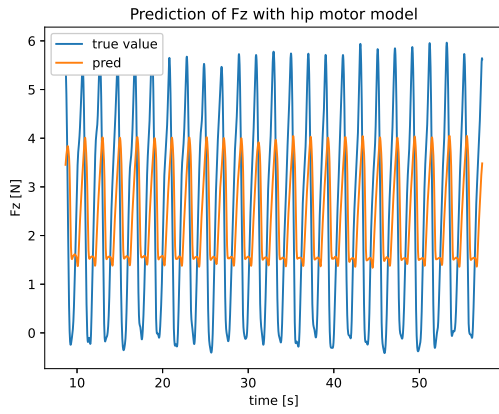


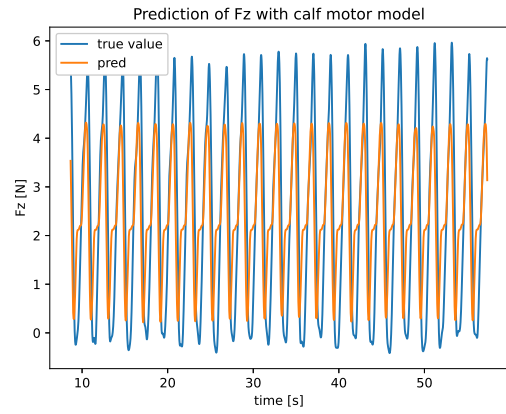
FIGURE A.2

Prediction of F_z with simple multiple linear regression model for an amplitude of 0.2 rad.

A.1.2 AMPLITUDE OF 0.35 RAD



(A) Prediction of F_z with hip motor model.



(B) Prediction of F_z with calf motor model.

FIGURE A.3

Prediction of F_z with simple linear regression model for an amplitude of 0.35 rad.

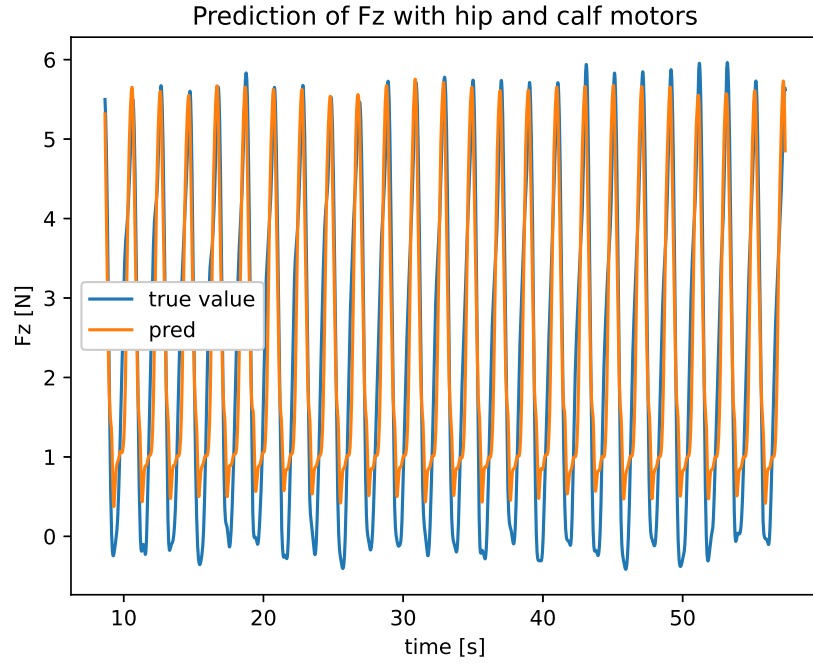
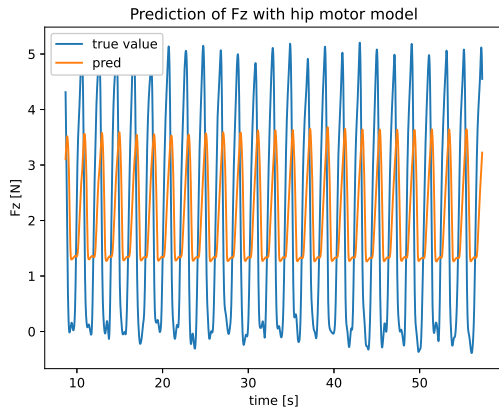


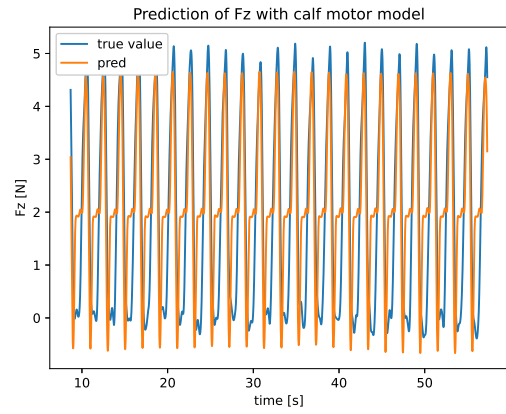
FIGURE A.4

Prediction of F_z with simple multiple linear regression model for an amplitude of 0.35 rad.

A.1.3 AMPLITUDE OF 0.5 RAD



(A) Prediction of F_z with hip motor model.



(B) Prediction of F_z with calf motor model.

FIGURE A.5

Prediction of F_z with simple linear regression model for an amplitude of 0.5 rad.

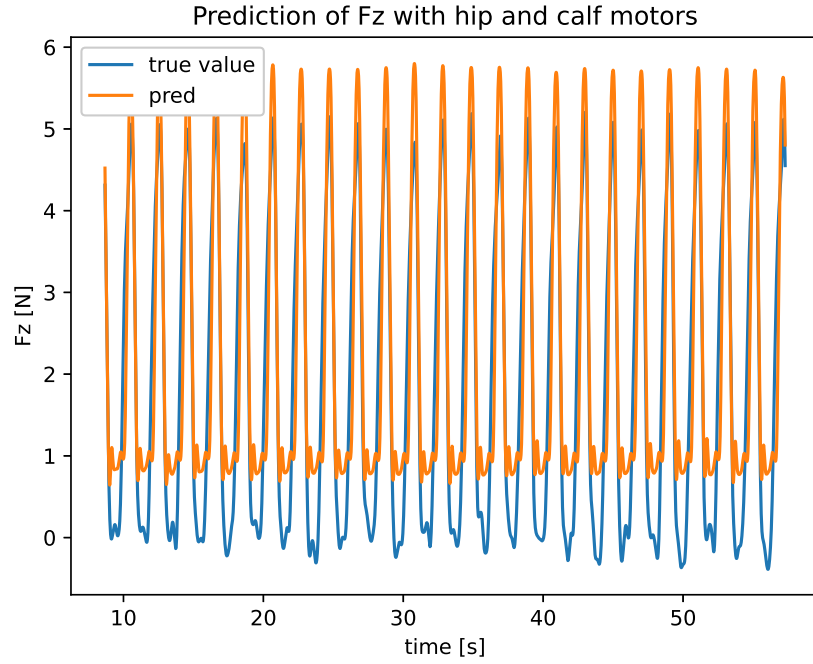
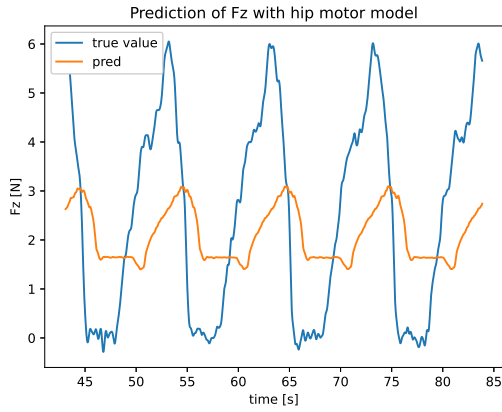


FIGURE A.6

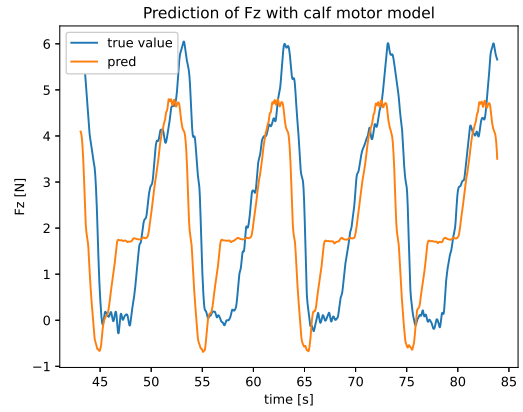
Prediction of Fz with simple multiple linear regression model for an amplitude of 0.5 rad.

A.2 INFLUENCE OF FREQUENCY - PLOTS

A.2.1 FREQUENCY OF 0.1 Hz



(A) Prediction of Fz with hip motor model.



(B) Prediction of Fz with calf motor model.

FIGURE A.7

Prediction of Fz with simple linear regression model for a frequency of 0.1 Hz.

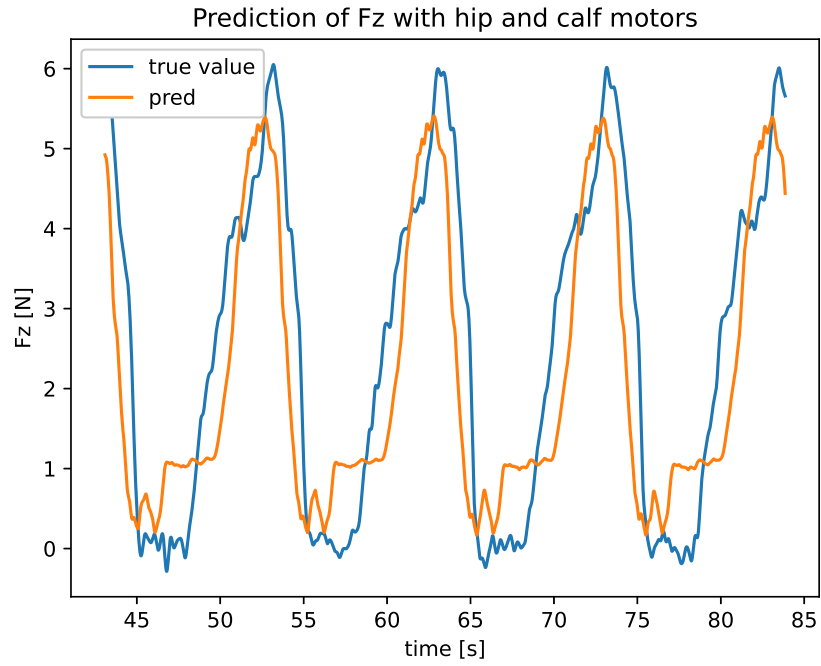
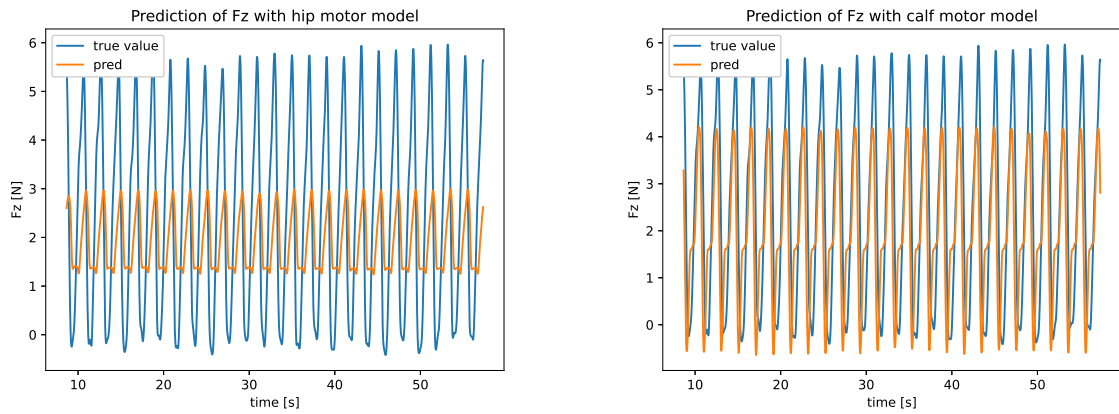


FIGURE A.8
Prediction of Fz with simple multiple linear regression model for an frequency of 0.1 Hz.

A.2.2 FREQUENCY OF 0.5 HZ



(A) Prediction of Fz with hip motor model.

(B) Prediction of Fz with calf motor model.

FIGURE A.9
Prediction of Fz with simple linear regression model for a frequency of 0.5 Hz.

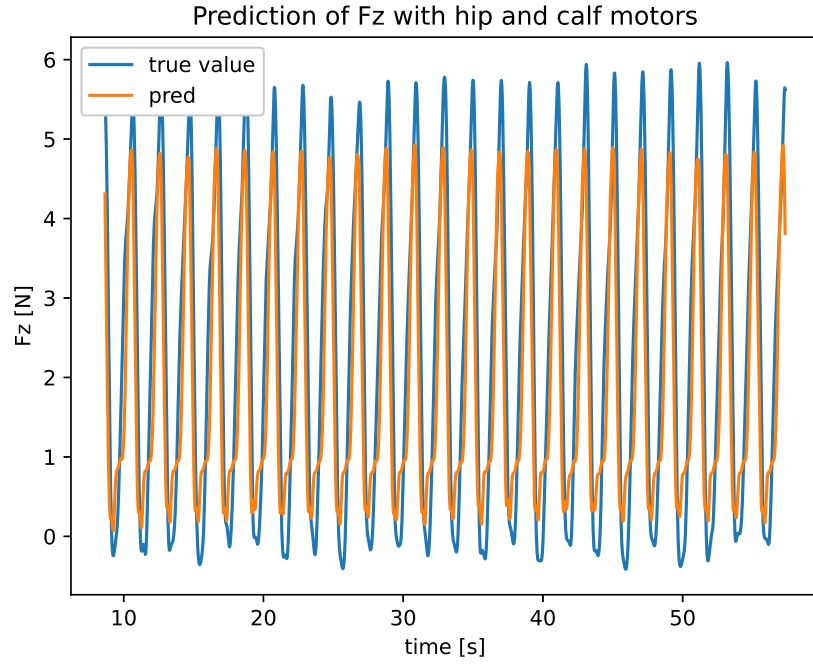
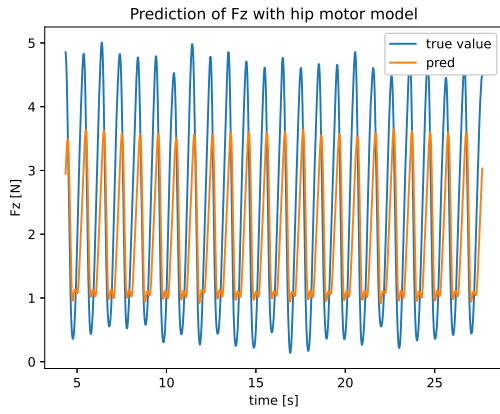


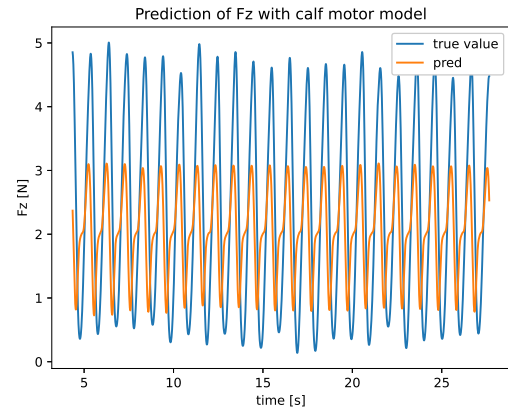
FIGURE A.10

Prediction of Fz with simple multiple linear regression model for an frequency of 0.5 Hz.

A.2.3 FREQUENCY OF 1.0 Hz



(A) Prediction of Fz with hip motor model.



(B) Prediction of Fz with calf motor model.

FIGURE A.11

Prediction of Fz with simple linear regression model for a frequency of 0.1 Hz.

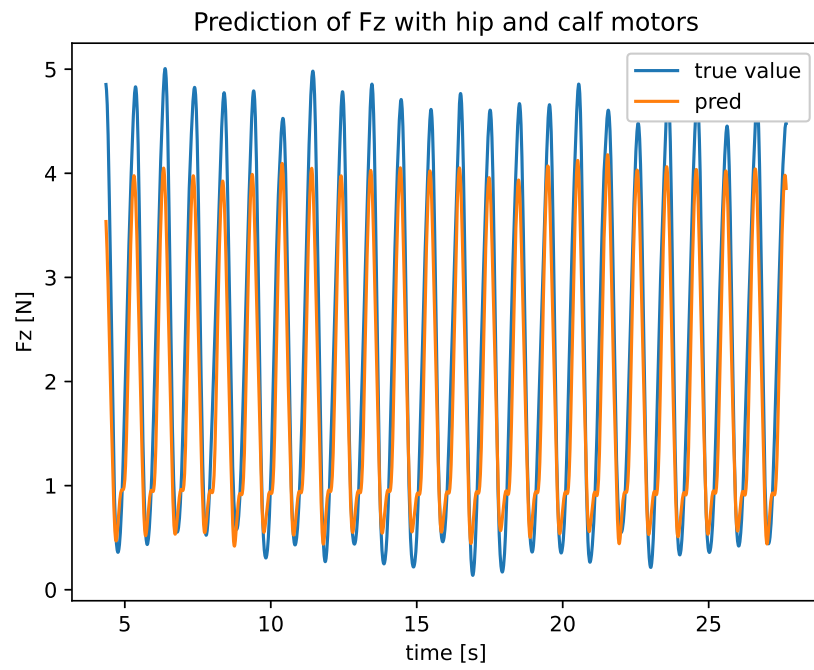


FIGURE A.12

Prediction of Fz with simple multiple linear regression model for an frequency of 1.0 Hz.